JÖNKÖPING UNIVERSITY

School of Engineering

Data Science: Assignment – 2

Submitted by

Group number: 9

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1. INTRODUCTION

Machine learning is often used to make predictions in real-life situations where the outcomes can be either categories or numbers. This assignment looks at how supervised learning works by focusing on two main types of tasks: classification and regression. The goal is to understand how to evaluate, compare, and improve machine learning models to make better predictions. The study uses two different datasets. One dataset is used for a classification task, where the goal is to predict whether a person's income is more than $50,000 a year, using the Adult Census Income dataset. The other dataset is for a regression task, where the task is to predict housing prices based on factors like location, income levels, and other social and economic features, using the California Housing dataset. These datasets help us explore different ways of building and comparing models. To reach the goals of this project, we followed a series of steps, including cleaning and preparing the data, selecting useful features, and training several machine learning models. At least three models for each task were tested using a method called 10-fold cross-validation. Their performance was compared using measurement tools like Mean Squared Error (MSE) and the coefficient of determination (R^2). After reviewing the results, the best-performing model was chosen and further improved by adjusting an important setting called a hyper parameter. Finally, the optimized model was trained on the full dataset and saved so it can be used for future predictions. The entire analysis was done using the KNIME Analytics Platform.

2. CLASSIFICATION

Classification helps create models that sort observations into groups or categories. This allows us to make choices or guesses based on patterns we learn from past data. In this project, we used classification methods to predict if someone's yearly income is more than $50,000. The information we used came from the Adult Census Income dataset, which includes details about people's age, job, education, and other related factors. This specific task has only two possible outcomes: either the person's income is $50,000 or less, or it's more than $50,000.

2.1 DATA PRE-PROCESSING

Data pre-processing was carried out in KNIME to prepare the Adult Census Income dataset for classification. The data was loaded using a CSV Reader, followed by String Manipulation nodes to clean and standardize text features. Irrelevant columns were removed using a Column Filter. Missing values marked with "?" were handled using the Missing Value node by replacing them with the most frequent values. Categorical features were converted to numerical form using the Category to Number node, and numerical features were scaled using a Normalizer to ensure consistency across features.

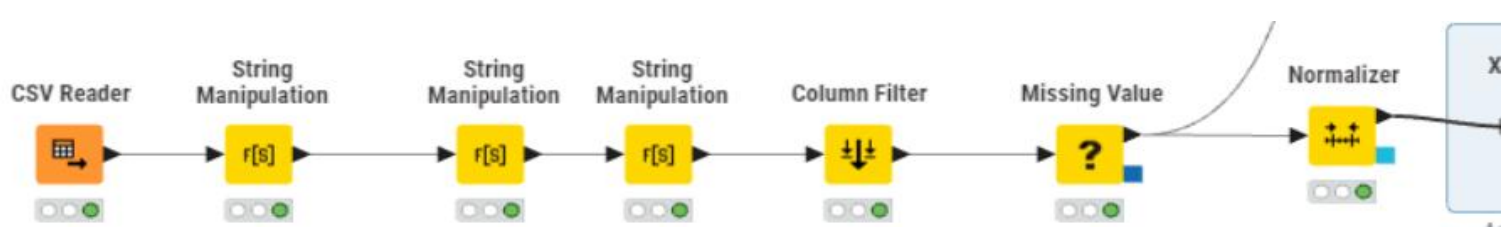


Figure 1: KNIME Workflow – Data Preprocessing

2.2 DATA ANALYSIS

The box plot above shows how age relates to income. It shows that the middle age of people making more than $50K is higher than those making $50K or less. Both groups also have some people who are much older, which means there's a lot of variation in income for older individuals. This suggests that age is an important factor when predicting income, as higher incomes tend to come from older age groups.

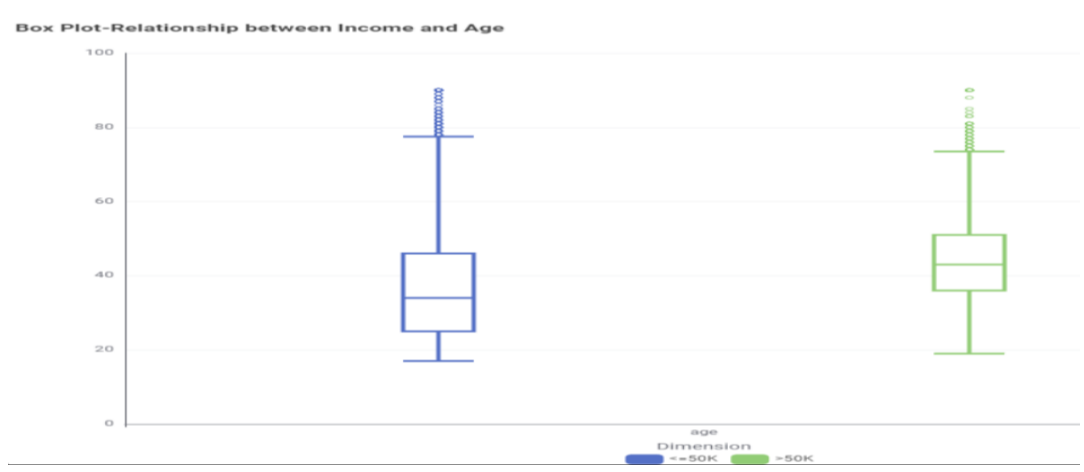


Figure 2: Box Plot-Relationship between Income and Age

Training data (Table)

Rows: 43957 | Columns: 14

#	RowID	age	workclass	fnlwgt	education	educatio...	marital-s...	occupation	relations...	g
1	Row0	0.11	0	0.145	0	0.4	0	0	0	0
2	Row1	0.288	0	0.052	0.067	0.533	0.167	0.071	0.2	0
3	Row3	0.37	0	0.1	0.2	0.6	0.167	0	0.2	0
4	Row4	0.014	0.25	0.062	0.2	0.6	0	0.214	0	1
5	Row6	0.164	0.25	0.145	0.067	0.533	0	0.214	0.6	0
6	Row7	0.63	0.375	0.062	0.333	0.933	0.167	0.357	0.2	0
7	Row8	0.096	0	0.242	0.2	0.6	0	0.286	0.6	1
8	Row9	0.521	0	0.063	0.4	0.2	0.167	0.429	0.2	0
9	Row10	0.658	0	0.116	0.067	0.533	0.167	0	0.2	0
10	Row12	0.123	0	0.047	0.067	0.533	0	0.5	0.4	1
11	Row13	0.562	0.25	0.195	0.067	0.533	0.167	0.214	0.2	0
12	Row14	0.425	0	0.181	0.067	0.533	0.167	0	0.2	0
13	Row15	0.356	0	0.226	0.533	0.867	0.167	0.571	0.2	0
14	Row16	0.041	0.625	0.292	0.2	0.6	0	0.286	0	0
15	Row17	0.356	0	0.079	0.067	0.533	0.167	0.5	0.8	1
16	Row18	0.274	0	0.033	0.067	0.533	0.333	0	0.6	1
17	Row19	0.315	0	0.049	0.6	1	0.167	0.357	0.2	0
18	Row20	0.233	0	0.065	0.467	0.8	0.167	0.643	0.2	0
19	Row21	0.233	0	0.153	0.2	0.6	0	0.286	0	1
20	Row22	0.753	0.25	0.081	0.4	0.2	0.5	0.214	0.4	1
21	Row23	0.11	0	0.141	0.467	0.8	0	0.357	0.4	0
22	Row24	0.11	0	0.131	0.467	0.8	0.167	0.357	0.2	0
23	Row25	0.384	0.375	0.285	0.067	0.533	0.167	0.429	0.2	0
24	Row26	0.068	0	0.152	0.067	0.533	0	0.5	0	0
25	Row27	0.082	0	0.083	0.067	0.533	0.667	0	0.6	0

Figure 3: values taken under classification

2.3 CLASSIFIERS: SELECTION AND COMPARISON

2.3.1 SELECTION

1. Random Forest Classifier

The Random Forest classifier is a type of machine learning that uses many decision trees to make predictions. In the process, the Random Forest Learner node is used to teach the model using the training data, and the Random Forest Predictor node is used to make predictions on new, unseen data.

2. k-Nearest Neighbors (k-NN)

The k-Nearest Neighbors classifier is a method that uses the nearest data points to decide the class of a new point.t looks at the majority class among the k closest neighbors. In the workflow, the k-Nearest Neighbor node is used for making predictions. Since this method depends on distance, the data was normalized during the prep step.

3. Gradient Boosted Trees

Gradient Boosted Trees is another method that uses multiple decision trees built one after another. Each new tree tries to fix the mistakes made by the previous ones. In the workflow, the Gradient Boosted Trees Learner is used to train the model, and the Gradient Boosted Trees Predictor is used to make predictions.

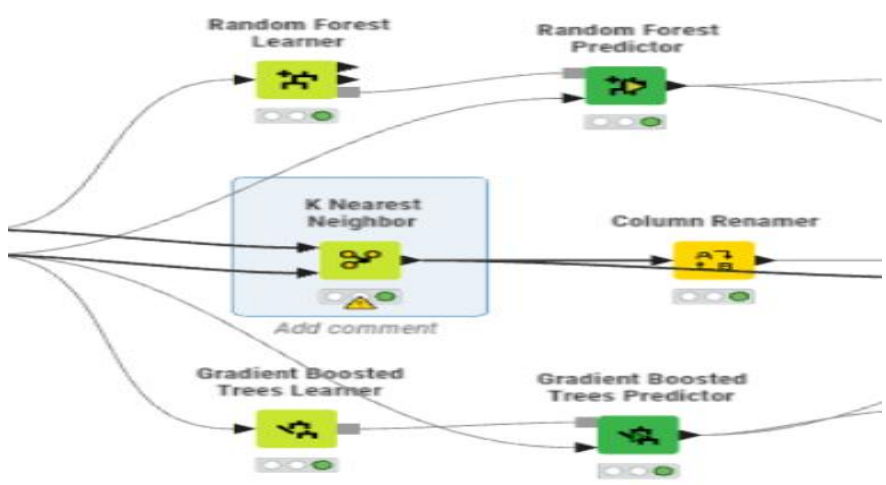


Figure 4: Machine Learning Classifiers

2.3.2 COMPARISON

For comparing the different classifiers 10-fold cross validation method is used. It evaluated the performance of machine learning models and making the efficient use of dataset. This method splits the dataset into 10 equal parts (folds). The model is trained and evaluated 10 times, each time using a different fold as the test set and the remaining 9 folds as the training set.

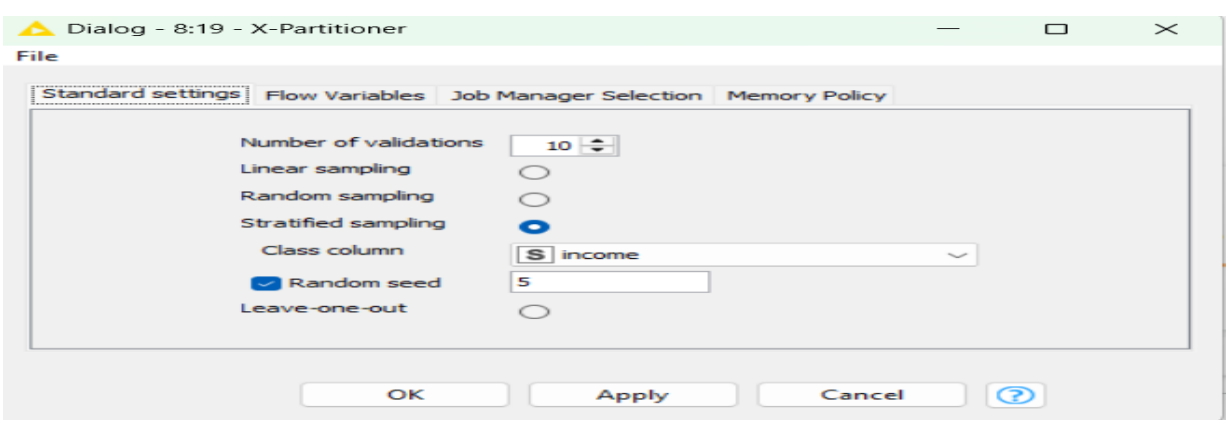


Figure 5: 10-fold cross validation method

To compare, two main measures, Accuracy and Area Under Curve (AUC), are used because they are the reliable at showing how well a model can classify things and clearly tell apart different categories. Looking at the table below, it's clear that the Accuracy Random Forest Classifier is more accurate than the others.

Model	Accuracy	**AUC**
Random Forest Classifier	Accuracy Number (Float) 0.865	0.870
k-Nearest Neighbors (k-NN)	Accuracy Number (Float) 0.811	0.774
Gradient Boosted Trees	Accuracy Number (Float) 0.835	0.842

Figure 6: Comparison Table

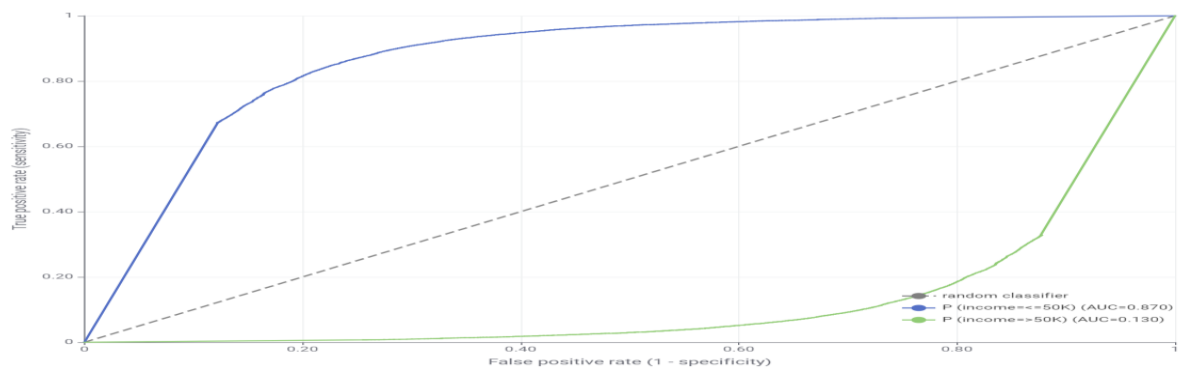


Figure 7: ROC curve for Random Forest

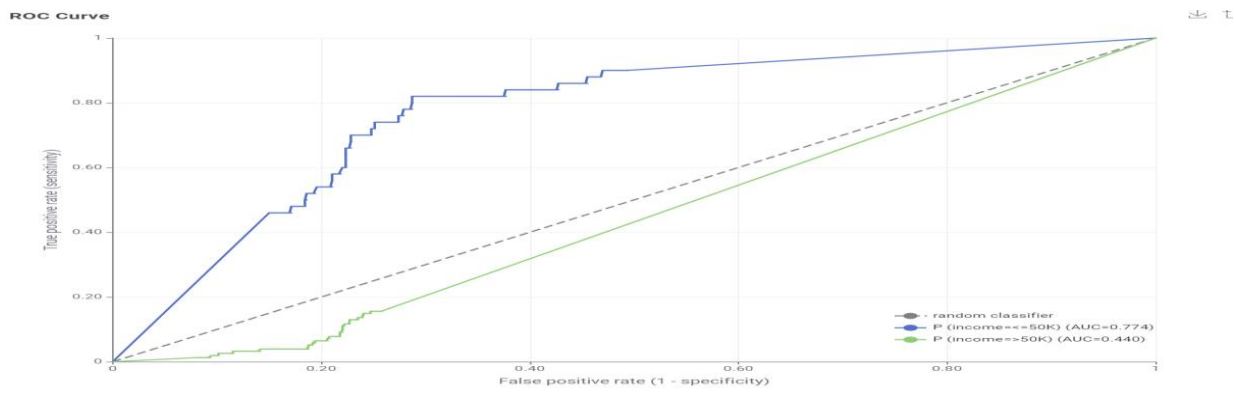


Figure 8: ROC curve for k-Nearest Neighbors

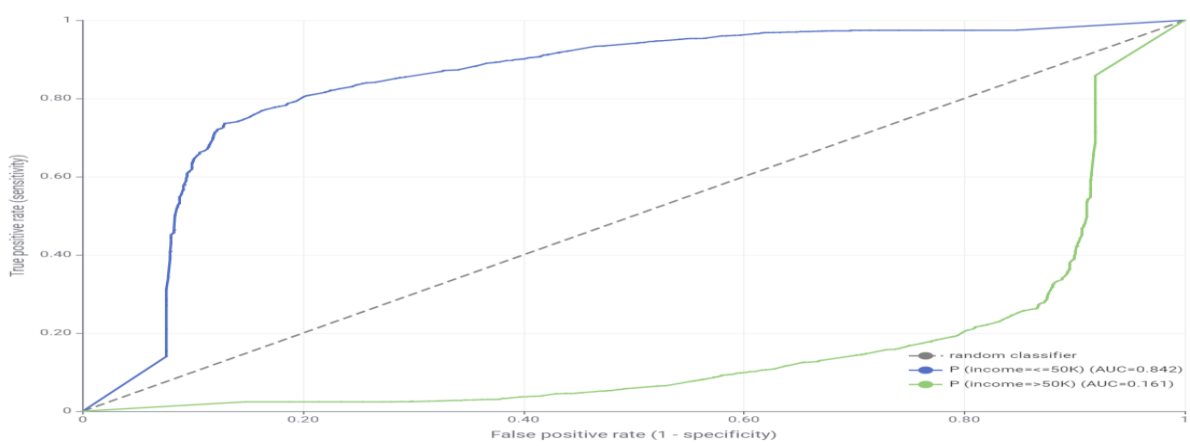


Figure 9: ROC curve for Gradient Boosted Trees

The bar graph illustrates how well three different classification models perform, the findings reveal that the Random Forest model demonstrates superior accuracy compared to both Gradient Boosted Trees and k-Nearest Neighbors, which exhibits the least efficiency in classification tasks

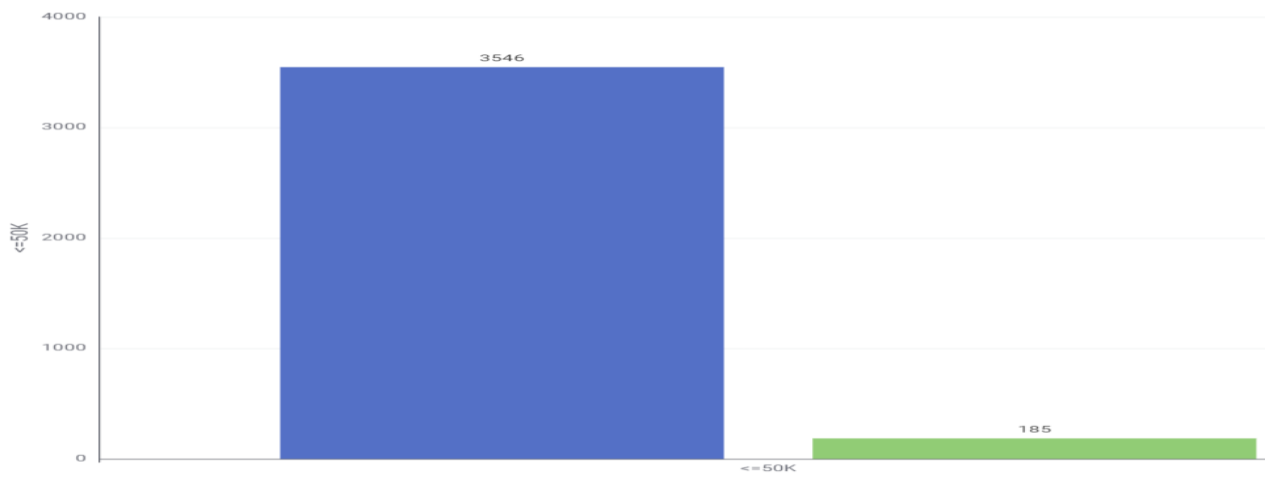


Figure 10: class distribution of income in Random Forest

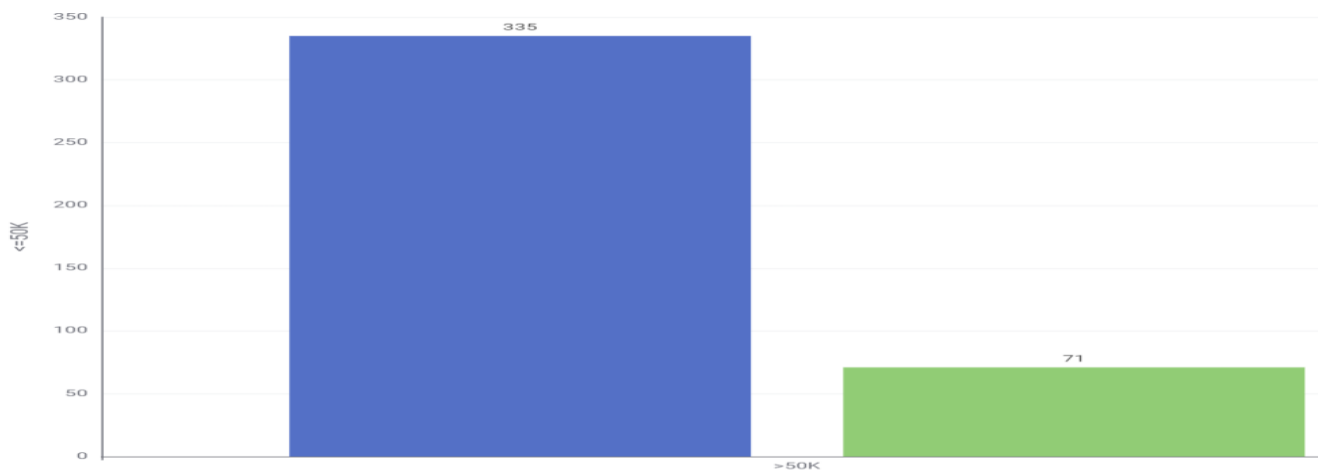


Figure 11: class distribution of income in k-Nearest Neighbors (k-NN)

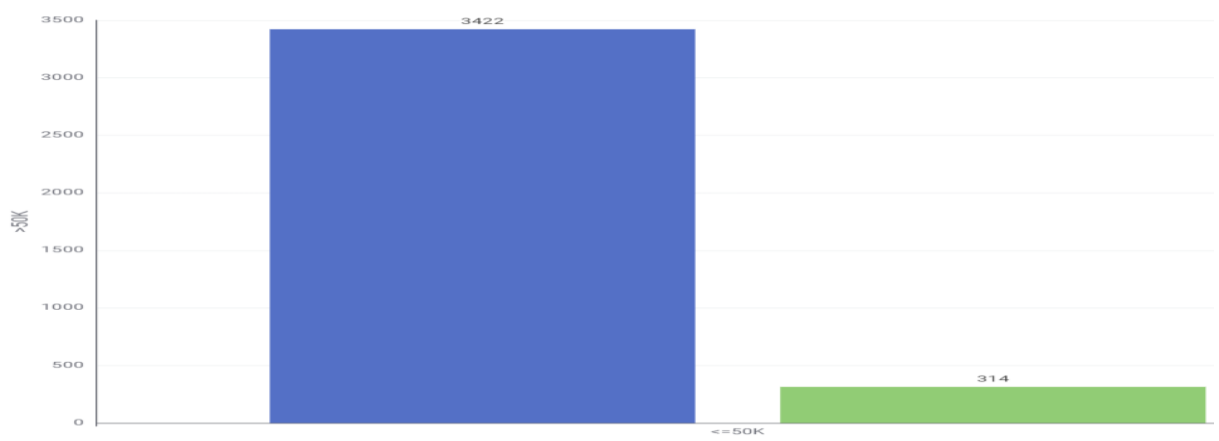


Figure 12: class distribution of income in Gradient Boosted Trees

2.4 MODEL OPTIMIZATION

It is an important step in machine learning task as it can enhance the performance of a model. We have used Parameter optimization loop to get a structured approach to explore various combinations of parameters. We have used Parameter optimization only for the Random Forest Classifier as it gave us better model when compared to other models. Random Forest Classifier Learner allow us to change many parameters in KNIME like the max number of levels, no of models, node size, child size etc. Hence, we can say that the tree depth in a Random Forest Classifier is a crucial hyper parameter that influences the trade-off between model complexities.

Tree Depth	Accuracy	AUC
5	0.833	0.790
10	0.865	0.832
15	0.865	0.832

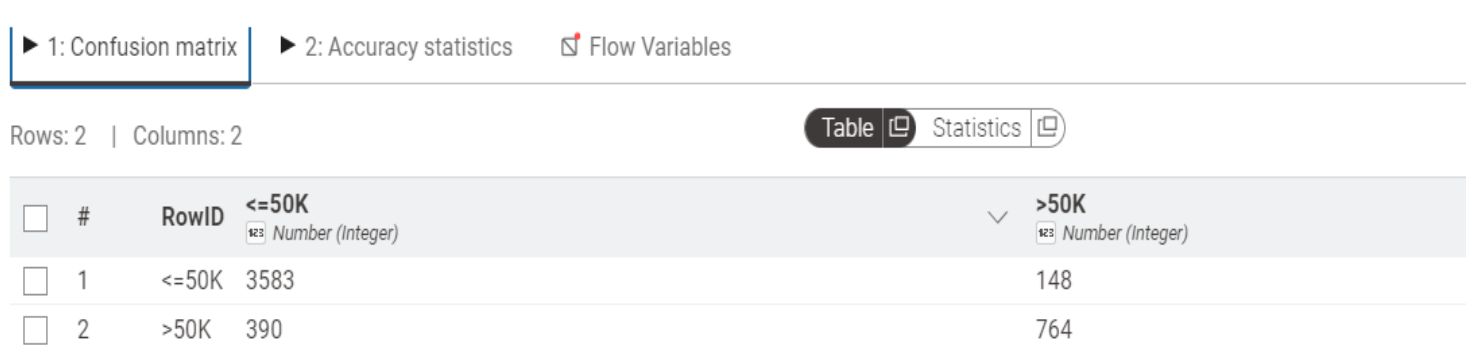
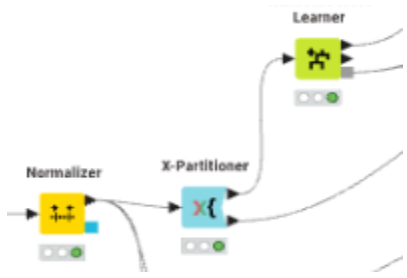


Figure 13: Confusion Matrix after tuning

2.5 MODEL TRAINING AND SAVING

The most performing model from the analysis which is Random Forest Classifier selected, optimized, and retrained using the complete dataset to maximize performance. The finalized model is saved using the Model

Writer node in KNIME. It can later be reused or deployed in another workflow by loading it with the Model Reader node.



Filtered table (Table)

Rows: 13189 | Columns: 9

#	RowID	age	workclass	education	marital-s...	occupation	relations...	gender	native-co...	in
1	Row0	0.164	Unknown	HS-grad	Never-married	Unknown	Unmarried	Male	Unknown	<=50K
2	Row1	0.26	Federal-gov	Bachelors	Married-civ-spo	Federal-gov	Husband	Male	Federal-gov	<=50K
3	Row2	0.233	Private	Bachelors	Married-civ-spo	Private	Husband	Male	Private	>50K
4	Row3	0.753	Unknown	7th-8th	Divorced	Unknown	Not-in-family	Female	Unknown	<=50K
5	Row4	0.397	State-gov	Some-college	Married-civ-spo	State-gov	Husband	Male	State-gov	>50K
6	Row5	0.301	Private	7th-8th	Married-civ-spo	Private	Husband	Male	Private	<=50K
7	Row6	0.479	Private	11th	Separated	Private	Not-in-family	Female	Private	<=50K
8	Row7	0.301	Private	HS-grad	Divorced	Private	Own-child	Male	Private	<=50K
9	Row8	0.288	Private	9th	Married-spouse	Private	Not-in-family	Male	Private	<=50K
10	Row9	0.397	Private	Assoc-voc	Married-civ-spo	Private	Husband	Male	Private	<=50K
11	Row10	0.329	Private	HS-grad	Married-civ-spo	Private	Husband	Male	Private	<=50K
12	Row11	0.452	Private	HS-grad	Married-civ-spo	Private	Husband	Male	Private	<=50K
13	Row13	0.575	Federal-gov	Bachelors	Divorced	Federal-gov	Not-in-family	Male	Federal-gov	<=50K
14	Row14	0.26	Private	10th	Separated	Private	Not-in-family	Female	Private	<=50K
15	Row15	0.329	Local-gov	HS-grad	Married-civ-spo	Local-gov	Husband	Male	Local-gov	<=50K
16	Row16	0.315	Private	Some-college	Married-civ-spo	Private	Husband	Male	Private	<=50K
17	Row17	0.356	Private	Bachelors	Divorced	Private	Unmarried	Female	Private	>50K
18	Row18	0.014	Private	Some-college	Never-married	Private	Own-child	Female	Private	<=50K
19	Row19	0.644	Unknown	Some-college	Married-civ-spo	Unknown	Husband	Male	Unknown	<=50K
20	Row22	0.151	Self-emp-inc	Bachelors	Married-civ-spo	Self-emp-inc	Husband	Male	Self-emp-inc	<=50K
21	Row23	0.233	Self-emp-inc	Bachelors	Married-civ-spo	Self-emp-inc	Husband	Male	Self-emp-inc	<=50K
22	Row24	0.11	Private	Some-college	Never-married	Private	Own-child	Female	Private	<=50K
23	Row25	0.342	Private	HS-grad	Never-married	Private	Unmarried	Female	Private	<=50K
24	Row27	0.096	Private	Bachelors	Married-civ-spo	Private	Husband	Male	Private	>50K
25	Row29	0.562	Private	10th	Married-civ-spo	Private	Husband	Male	Private	>50K

Figure 14: Table with prediction for entire dataset

3 REGRESSION

Regression is a kind of machine learning method that helps predict numbers. In this task, we will build a model that can estimate the average house price in a neighbourhood. This average price is the middle value of all house prices in that area.

3.1 DATA PRE-PROCESSING

Before training the regression model, the dataset is prepared to improve data quality and consistency. Missing values marked as "?" are replaced with the average value of the corresponding attribute. All numerical features are then scaled to a common range using normalization, which helps the model learn more effectively. Finally, the dataset is split into training and testing subsets using the X-Partitioner node in KNIME, as shown in the workflow above.

Figure 15: Workflow for Feature Selection

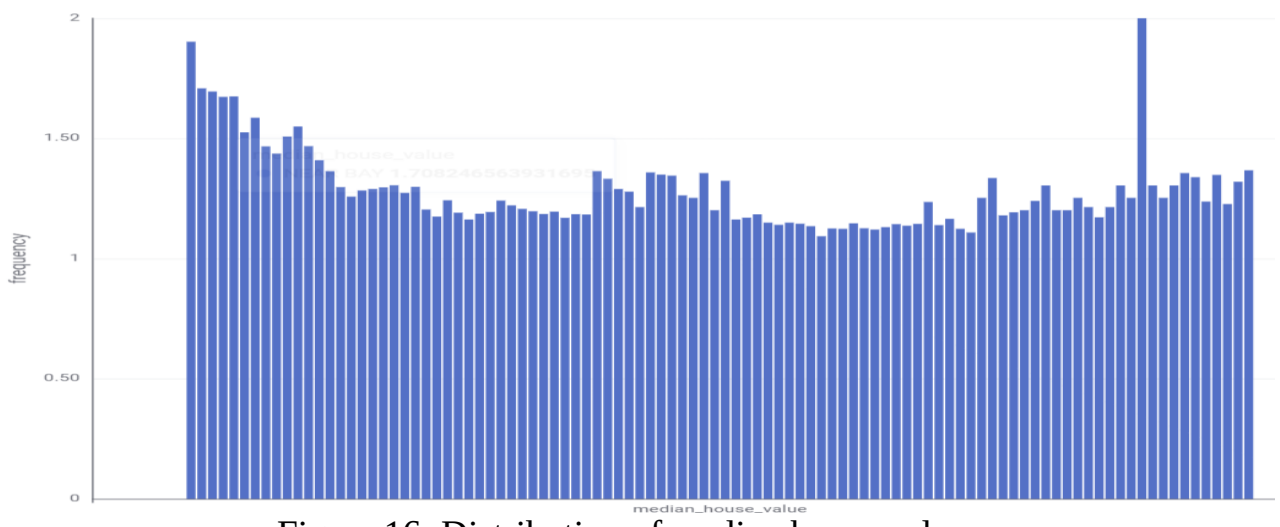


Figure 16: Distribution of median house values

The histogram shows the distribution of median house values across districts. Most homes fall within the 0.40 to 0.80 range, indicating that most neighbourhoods in the dataset have moderate housing prices.

3.2 REGRESSORS: SELECTION AND COMPARISON

3.2.1 MODEL SELECTION

For this regression task, three different learning methods were used to predict the median house value.

1. Random Forest Regression

Random Forest uses many decision trees to make a final prediction .Each tree is built using a random mix of data and features, which helps improve accuracy and prevent overfitting. This method works well with complex data.

2. Simple Regression Tree Learner

The Simple Regression Tree Learner is a supervised learning algorithm that predicts a continuous target variable by splitting the data into smaller groups based on feature values. It uses a tree structure with decision rules at internal nodes and numerical predictions at leaf nodes, typically the mean of the target values.

3. Gradient Boosted Trees Regression

Gradient Boosted Trees builds trees one after another. Each new tree fixes the mistakes made by the previous ones. This approach is great at finding non-linear patterns and how different features interact, which helps make better predictions.

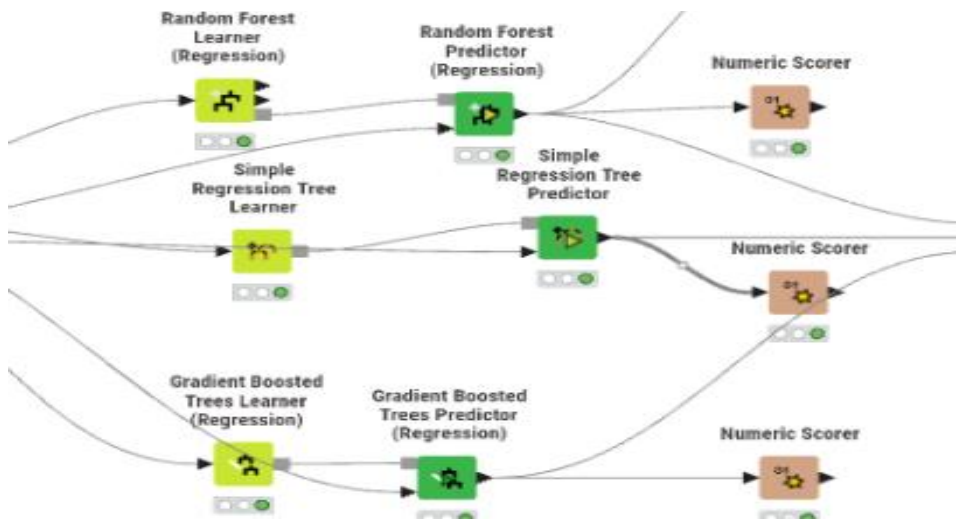


Figure 17: Machine Learning Regressors

3.2.2 COMPARISON

To check how well different regression models work, a 10-fold cross-validation method was used.

This helps evaluate machine learning models well while making the best use of the data. In this method, the data is split into 10 equal groups called folds. The model is trained and tested 10 times. Each time, one fold is used as the test group, and the other nine folds are used for training.

The models were measured using two things: Mean Squared Error (MSE) and R-squared (R^2).

A smaller MSE means the model predicts more accurately, and a higher R^2 means the model fits the data better. From the tables, it shows that the Random Forest regressor performs better than the other models.

Rows: 7 | Columns: 1

#	RowID	prediction (Random Forest Regression) Number (Float)
1	R^2	0.846
7	adjusted R^2	0.846
6	mean absolute percentage error	0.24
4	root mean squared error	0.093
2	mean absolute error	0.063
3	mean squared error	0.009
5	mean signed difference	0.002

Figure 18: Random Forest Table

#	RowID	prediction(simple Regression) Number (Float)
1	R^2	0.663
2	mean absolute error	0.088
3	mean squared error	0.019
4	root mean squared error	0.139
5	mean signed difference	-0.001
6	mean absolute percentage error	0.256
7	adjusted R^2	0.663

S

Figure 19: Simple Regression Table

Rows: 7 | Columns: 1

#	RowID	Prediction (Gradient Boosted Trees) Number (Float)
1	R^2	0.806
2	mean absolute error	0.07
3	mean squared error	0.011
4	root mean squared error	0.105
5	mean signed difference	-0.003
6	mean absolute percentage error	0.048
7	adjusted R^2	0.806

Figure 20: Gradient Boosted Table

Model	R^2	MSE
Random Forest Regression	0.846	0.009
simple Regression	0.663	0.019
Gradient Boosted Trees Regression	0.806	0.011

Figure 21: Comparison Table

The regression line illustrates the relationship between the predicted and actual values for the Regression model, showing a moderate fit between the variables and from the analysis , Random Forest Regression achieves superior performance, as indicated by its higher R^2 value and lower MSE.

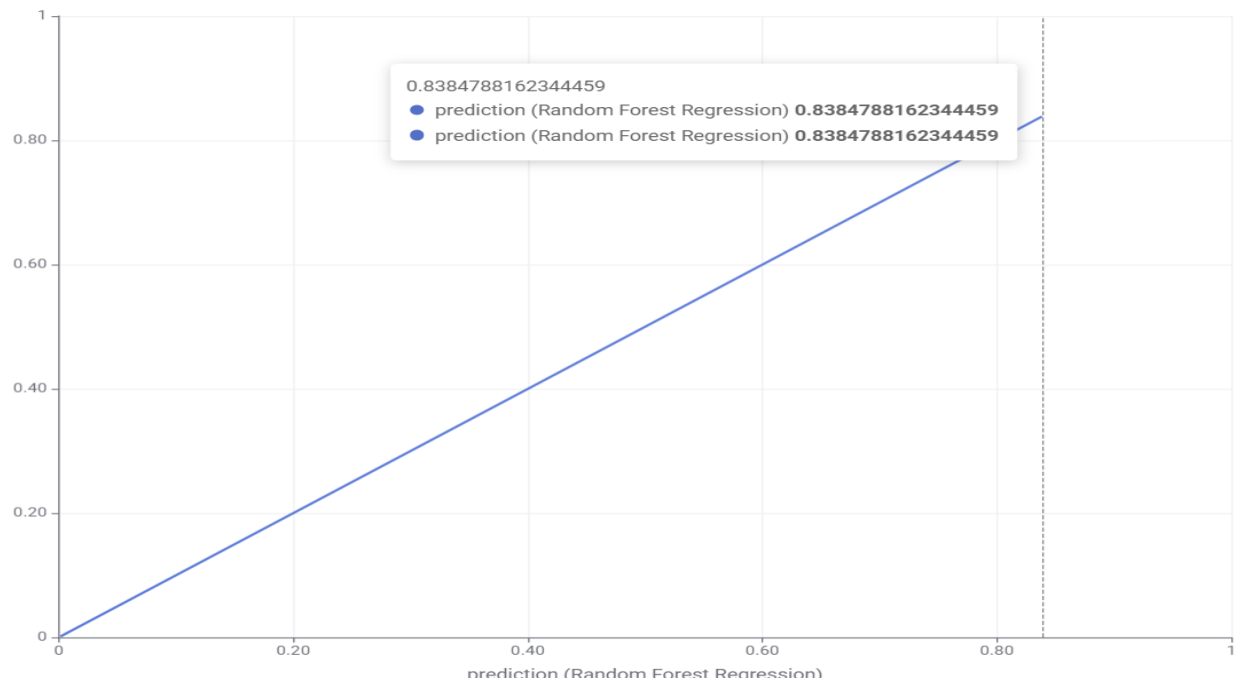


Figure 22: Predicted vs Actual Values for Random Forest Regression

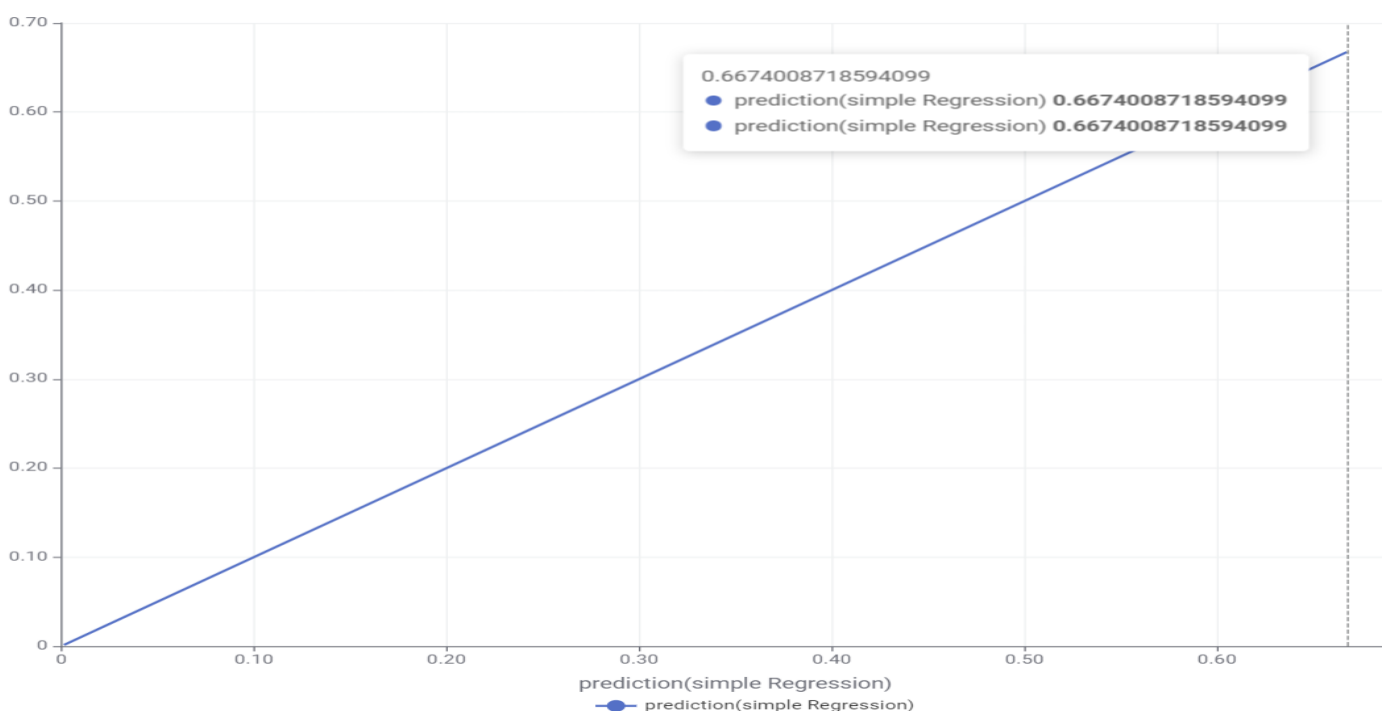


Figure 23: Predicted vs Actual Values for Simple Regression

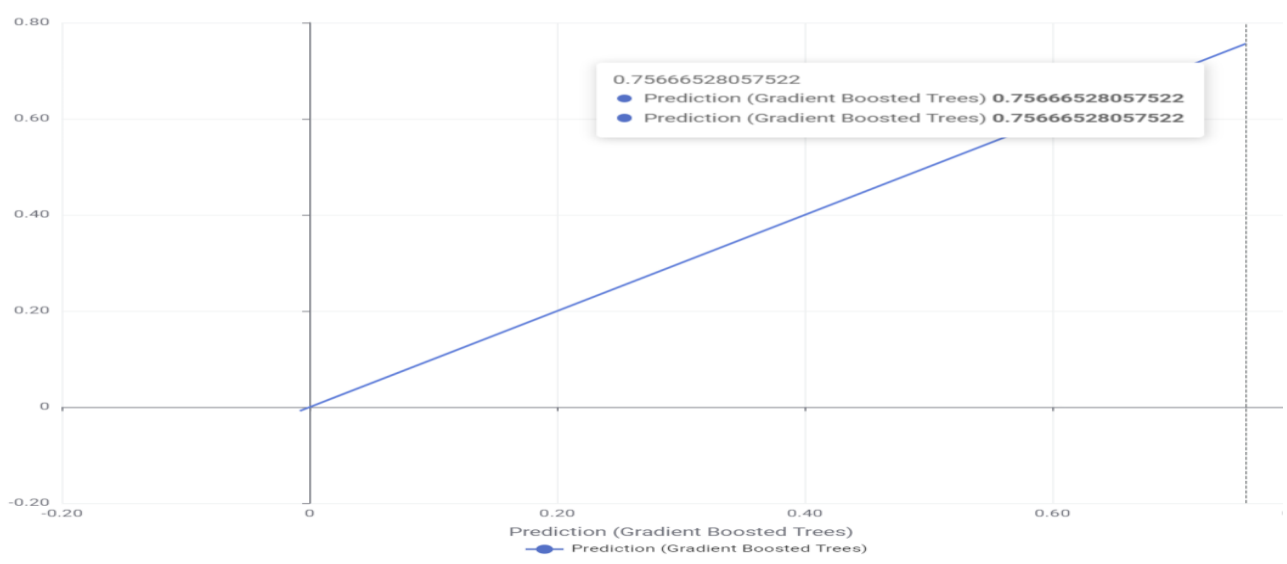


Figure 24: Predicted vs Actual Values for Gradient Boosted Trees

3.3 MODEL OPTIMIZATION

To optimize the Random Forest Regression model, GridSearchCV is used to extensively tune the number of estimators or trees .This leads to an enhanced accuracy of a more generalized model, as it tries to find better prediction performance on unseen data by doing a systematic analysis of the given predefined parameter. Higher R^2 means the model fits the data better. From the above tables, it shows that the Random Forest regressor performs better than the other models.

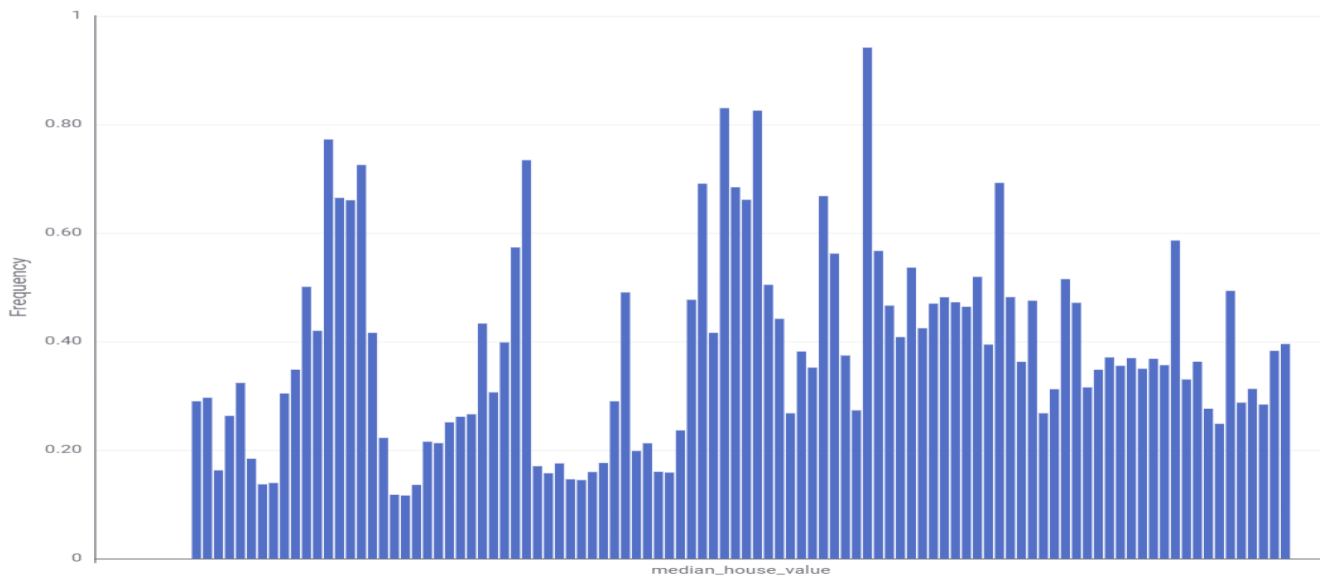


Figure 25: Random Forest with more optimized graph

3.4 MODEL TRAINING AND SAVING

Similar to Classification, the best model is now trained on the whole dataset and then is saved so that it can be used for further reference.

#	RowID	housing_median_age Number (Float)	total_bedrooms Number (Float)	population Number (Float)	median_income Number (Float)	median_house_value Number (Float)	Prediction (median_hous... Number (Float)	Prediction (median_hous... Number (Float)
1	Row0	1	0.029	0.014	0.466	0.695	0.877	0.009
2	Row1	1	0.076	0.031	0.218	0.586	0.508	0.008
3	Row2	1	0.106	0.032	0.181	0.467	0.491	0.016
4	Row3	1	0.073	0.031	0.178	0.409	0.491	0.014
5	Row4	1	0.046	0.019	0.145	0.305	0.316	0.014
6	Row5	1	0.084	0.028	0.084	0.204	0.351	0.017
7	Row6	0.98	0.089	0.035	0.154	0.195	0.368	0.011
8	Row9	1	0.045	0.017	0.069	0.149	0.227	0.011
9	Row10	0.745	0.037	0.011	0.181	0.459	0.329	0.013
10	Row11	0.765	0.104	0.045	0.487	0.818	0.754	0.015
11	Row12	0.608	0.041	0.018	0.539	0.724	0.807	0.019
12	Row13	0.745	0.083	0.035	0.402	0.717	0.715	0.01
13	Row14	1	0.052	0.023	0.452	0.652	0.733	0.023
14	Row15	0.549	0.134	0.037	0.062	0.369	0.348	0.031
15	Row16	1	0.061	0.03	0.116	0.255	0.245	0.01
16	Row18	0.902	0.061	0.035	0.112	0.166	0.209	0.002
17	Row19	0.745	0.02	0.011	0.076	0.201	0.214	0.012
18	Row20	1	0.033	0.028	0.174	0.136	0.292	0.016
19	Row21	1	0.029	0.017	0.179	0.227	0.259	0.01
20	Row22	0.667	0.063	0.032	0.164	0.266	0.276	0.008
21	Row24	0.824	0.077	0.031	0.307	0.532	0.511	0.02
22	Row25	0.529	0.118	0.049	0.168	0.259	0.303	0.015
23	Row26	0.941	0.038	0.022	0.176	0.253	0.263	0.01
24	Row27	0.765	0.018	0.015	0.159	0.206	0.229	0.015
25	Row28	0.804	0.039	0.025	0.09	0.159	0.162	0.002
26	Row31	1	0.064	0.031	0.263	0.308	0.424	0.012

Figure 26: Table with prediction for dataset

4 CONCLUSION

The research effectively utilized machine learning methods for tackling classification and regression tasks by employing datasets including the “Adult Income” and “Housing” databases. A structured approach was employed where initial steps involved cleaning and organizing raw information before diving into descriptive statistics and comparisons among various algorithms. Among those evaluated, the Random Forest method demonstrated notably better performance across all assigned objectives.

With 0.865 classification accuracy and 0.846 R^2 and 0.009 MSE for regression, hyper parameter tuning also significantly improved model performance .Observations regarding improved outcomes post-hyper parameter adjustment underscore the significance of optimizing models for enhanced efficiency. In summary, the findings indicate that Random Forest stands as an effective and adaptable technique capable of managing substantial and intricate data sets frequently found in practical prediction tasks.